

EE708: Fundamentals of Data Science and Machine Intelligence

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Module 3A Regression Analysis and Modeling: Linear, Non-Linear, and Logistic

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Regression

- It is a statistical approach used to analyze the relationship between a dependent variable and one or more independent variables.
- The objective is to determine the most suitable function that characterizes the connection between these variables.
- Data for regression problems comes as a set of N input/output observation pairs

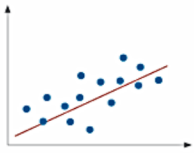
$$\{(\mathbf{x}_n, y_n)\}_{n=1}^N \triangleq (\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)$$

- $\mathbf{x}_n$: independent variable known as input/ regressor/ predictors/ covariates.
- Each dimension of the input is referred to as a *feature/explanatory variable*.
- y_n : dependent variable known as output/ response/prediction/estimation.
- In classification, the output values y_n are called *labels*, and all points sharing the same label value are referred to as a *class* of data.

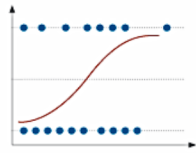
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Regression

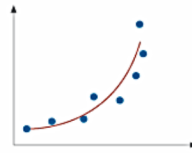
- Regression is performed to
 - produce a trend line or a curve that can be used to help visually summarize.
 - drive home a particular point about the data under study.
 - learn a model to precise predictions can be made regarding output values in the future.
- Types of regression



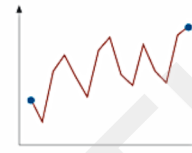
Linear regression: Predicts continuous output by modeling a straight-line relationship



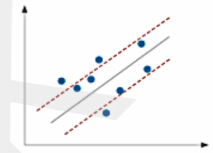
Logistic regression: Models the probability of binary outcomes



Polynomial regression: Capture nonlinear relationship by fitting polynomial curve



Time series regression: Predicts future values in a time-dependent data set.



Support vector regression: Approx. continuous function using a hyperplane that best fits the data.

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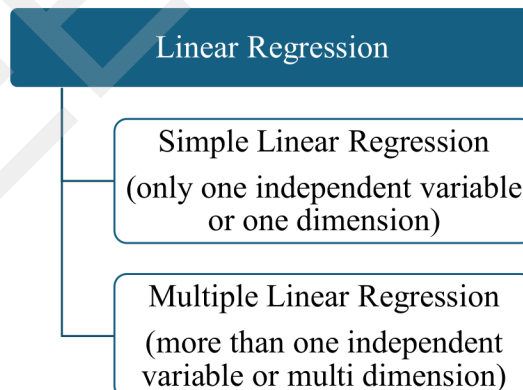
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Linear Regression

- Linear regression is a type of supervised learning algorithm.
- It computes the linear relationship between the dependent variable and one or more independent features by fitting a [linear equation](#) to observed data.



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Simple Linear Regression

- Fitting of a representative line to a set of input/output data points.
- For an input $\mathbf{x}_n \in \mathbb{R}^N$, it is a column vector of length N , given by

$$\mathbf{x}_n = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$$

- The relationship between input and output is

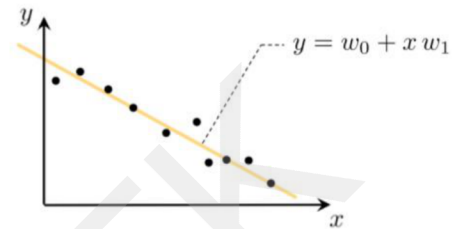
$$y_n \approx w_0 + w_1 x_n$$

- The linear equation is called the model $f_{w_0, w_1}(\mathbf{x}_n)$

$$f_{w_0, w_1}(\mathbf{x}_n) = w_0 + w_1 x_n$$

Where w_0, w_1 are the parameters known as

w_1 : regression coefficient; w_0 : bias/ intercept



Simple Linear regression one independent variable x_n

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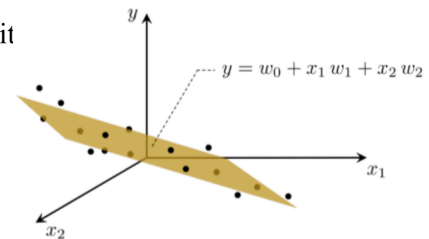
Multiple Linear Regression

- This involves more than one independent variable and one dependent variable.
 - With a bias and P associated slope weights to tune properly
- $$y_n \approx w_0 + x_{1,n} w_1 + x_{2,n} w_2 + \dots + x_{P,n} w_P, \quad n = 1, \dots, N$$
- The ' $\approx$ ' sign as it cannot be sure that all data lies completely on a single line.
 - Using notation $\dot{\mathbf{x}}$ to denote an input $\mathbf{x}$ with a 1 placed on top of it

$$\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_P \end{bmatrix}, \quad \dot{\mathbf{x}}_n = \begin{bmatrix} 1 \\ x_{1,n} \\ x_{2,n} \\ \vdots \\ x_{P,n} \end{bmatrix}$$

- Desired linear relationships is more compactly given by

$$y_n \approx f_w(\mathbf{x}_n) = \dot{\mathbf{x}}_n^T \mathbf{w}, \quad n = 1, \dots, N$$



Multiple Linear regression for $P = 2$

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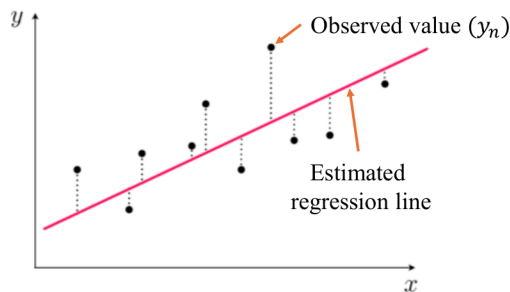
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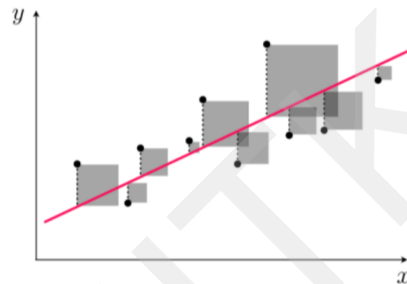
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Linear Regression: Least Squares Cost Function

- Find a weight vector $\mathbf{w}$ that
 - tightly holds each of N approximate equalities
 - Or the error between $\hat{\mathbf{x}}_n^T \mathbf{w}$ and y_n to be small
 - Or deviation of the observations from the true regression line is small



The best line and its errors



The best line and its squared errors

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Linear Regression: Least Squares Cost Function

- Point-wise cost that measures the error/ residual (ϵ_n) of a model for the point $\{\mathbf{x}_n, y_n\}$ is given by

$$g_n(\mathbf{w}) = \epsilon_n^2 = (\hat{\mathbf{x}}_n^T \mathbf{w} - y_n)^2$$
- For all N such values to be small, cost is averaged over dataset, forming a *least squares cost function* $g(\mathbf{w})$

$$g(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N g_n(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (\hat{\mathbf{x}}_n^T \mathbf{w} - y_n)^2$$

- It is a function of the weights $\mathbf{w}$ and data as well.
- The best fitting hyperplane is the one whose parameters minimize this error.

$$\min_{\mathbf{w}} g(\mathbf{w}) = \min_{\mathbf{w}} \frac{1}{N} \sum_{n=1}^N (\hat{\mathbf{x}}_n^T \mathbf{w} - y_n)^2$$

Objective: Find the parameters ' $\mathbf{w}$ ' that minimize the cost function.

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Linear Regression: Least Squares Estimators

Recollect: $y_n \approx w_0 + x_{1,n}w_1 + x_{2,n}w_2 + \dots + x_{P,n}w_P$

- Consider $P = 1$; then $g(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n^T \mathbf{w} - y_n)^2 = \frac{1}{N} \sum_{n=1}^N (w_0 + w_1 x_n - y_n)^2$ (a-b)²
- The least square estimators of w_0 and w_1 , say $\hat{w}_0$ and $\hat{w}_1$ must satisfy

$$\left. \frac{\partial g(\mathbf{w})}{\partial w_0} \right|_{\hat{w}_0, \hat{w}_1} = 0$$

$$\Rightarrow \frac{2}{N} \sum_{n=1}^N (w_0 + w_1 x_n - y_n) = 0$$

$$\Rightarrow N\hat{w}_0 + \hat{w}_1 \sum_{n=1}^N x_n = \sum_{n=1}^N y_n$$

$$\Rightarrow \boxed{\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x}}$$

$$\left. \frac{\partial g(\mathbf{w})}{\partial w_1} \right|_{\hat{w}_0, \hat{w}_1} = 0$$

$$\Rightarrow \frac{2}{N} \sum_{n=1}^N (w_0 + w_1 x_n - y_n) x_n = 0$$

$$\Rightarrow \hat{w}_0 \sum_{n=1}^N x_n + \hat{w}_1 \sum_{n=1}^N x_n^2 = \sum_{n=1}^N y_n x_n$$

$$\Rightarrow \boxed{\hat{w}_1 = \frac{\sum_{n=1}^N y_n x_n - N \bar{x} \bar{y}}{\sum_{n=1}^N x_n^2 - \frac{(N \bar{x})^2}{N}} = \frac{S_{xy}}{S_{xx}}}$$

Substitute $\hat{w}_0$

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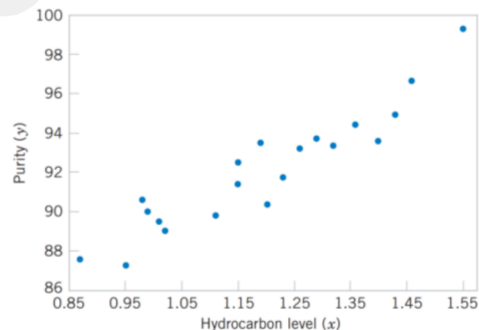
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Case Study: Oxygen Purity Dataset

Observation Number	Hydrocarbon Level x(%)	Purity y(%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33



Scatter diagram of oxygen purity versus hydrocarbon level

Objective: Fit a simple linear regression model to the oxygen purity data in Table

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Case Study: Oxygen Purity Dataset

- Solution: From Table

$$N = 20, \quad \sum_{n=1}^{20} x_n = 23.92, \quad \bar{x} = 1.196, \quad \sum_{n=1}^{20} y_n = 1843.21, \quad \bar{y} = 92.1605$$

$$\sum_{n=1}^{20} x_n^2 = 29.2892, \quad \sum_{n=1}^{20} y_n^2 = 170,044.5321, \quad \sum_{n=1}^{20} x_n y_n = 2214.6566$$

$$S_{xx} = 0.68088, \quad S_{xy} = 10.17744$$

The least square estimates:

$$\hat{w}_1 = \frac{S_{xy}}{S_{xx}} = 14.94748, \quad \hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x} = 74.2833$$

The fitted simple linear regression model:

$$\hat{y} = \hat{w}_0 + \hat{w}_1 x = 74.283 + 14.947x$$

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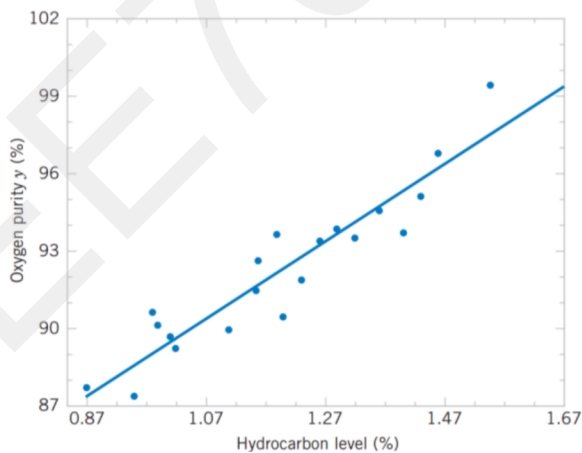
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Case Study: Oxygen Purity Dataset

Linear Reg.
 $y = w_0 + w_1 x$
 ↑
 We found this



Scatter plot of oxygen purity (y) versus hydrocarbon level (x) and regression model $\hat{y} = 74.283 + 14.947x$.

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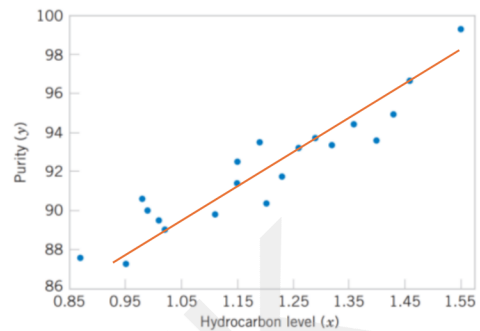
Linear Regression: Empirical Models

- The scatter diagram indicates
 - No simple curve will pass exactly through all the points.
 - The points are scattered randomly around the line.
- Hence y_n may be considered a random variable and related to x_n as

$$E(y_n|x_n) = w_0 + w_1 x_n$$

- For a fixed x_n the actual y_n is determined the linear model plus a random error term

$$y_n = w_0 + w_1 x_n + \epsilon$$



Scatter diagram of oxygen purity versus hydrocarbon level and the Regression Line

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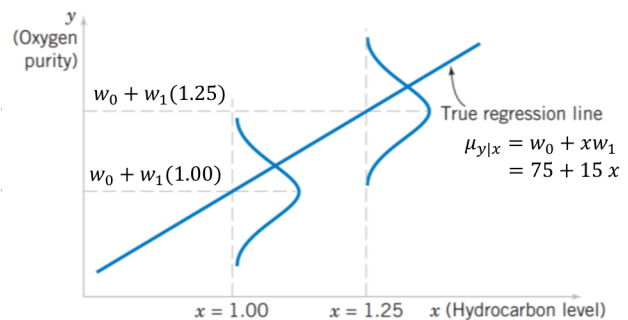
Linear Regression: Empirical Models

- Let the mean and variance of ϵ are $\mu = 0$ and σ^2 , then mean of y given x

$$\begin{aligned} E(y|x) &= E(w_0 + xw_1 + \epsilon) \\ &= w_0 + xw_1 + E(\epsilon) \\ &= w_0 + xw_1 \end{aligned}$$

- Variance of y given x

$$\begin{aligned} var(y|x) &= var(w_0 + xw_1 + \epsilon) \\ &= var(\epsilon) = \sigma^2 \end{aligned}$$



The distribution of y for given value of x for the oxygen purity-hydrocarbon data

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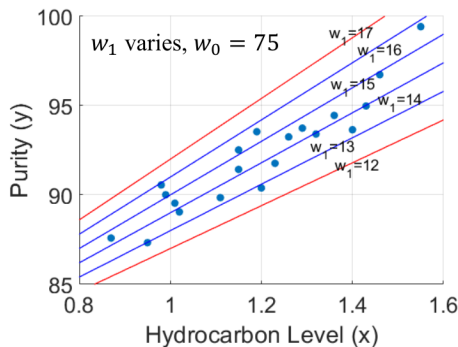
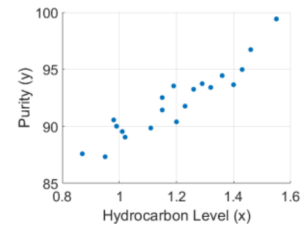
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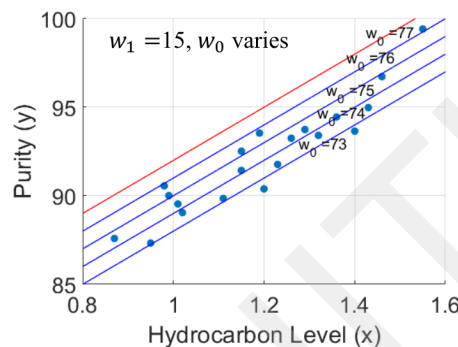
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Linear Regression: Role of w_0 and w_1

- Model: $f_{w_0, w_1}(x_n) = w_0 + w_1 x_n$; Parameters: w_0, w_1
- Cost function: $g(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (w_0 + w_1 x_n - y_n)^2$
- What do w_0 and w_1 do?



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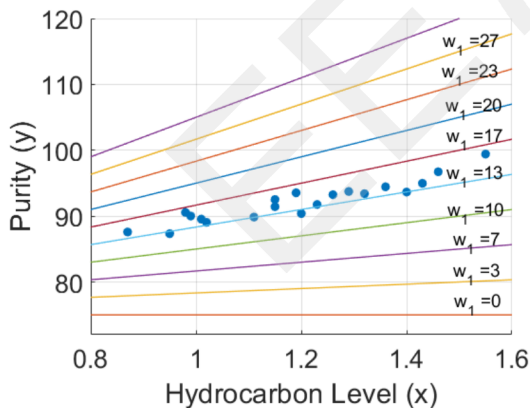
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Linear Regression: Gradient Cost

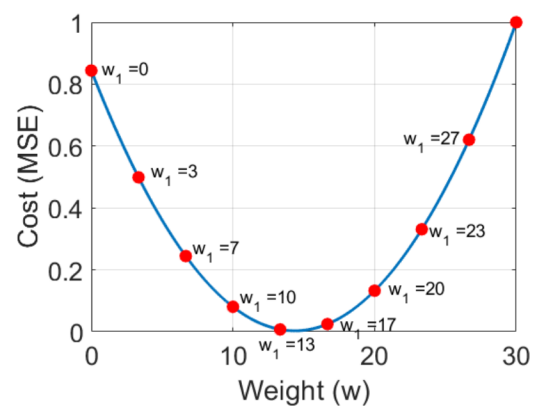
- Model: $f_{w_0, w_1}(x_n) = w_0 + w_1 x_n$;
- $g(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (f_{w_0, w_1}(x_n) - y_n)^2$

Model output by varying w_1 and $w_0 = 75$

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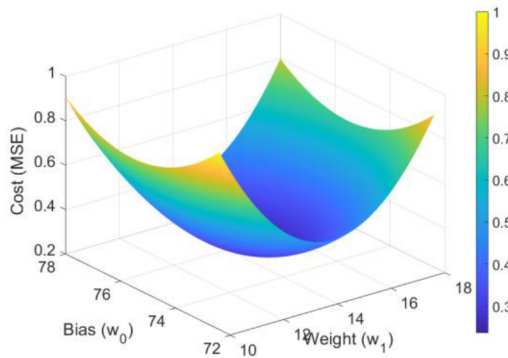
Cost (MSE) by varying w_1 and $w_0 = 75$

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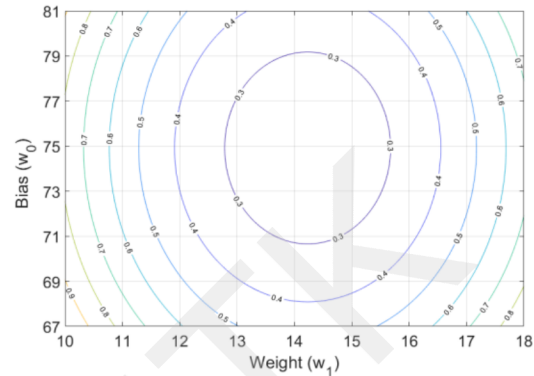
Linear Regression: Cost Function Visualization

• Model: $f_{w_0, w_1}(\mathbf{x}_n) = w_0 + w_1 \mathbf{x}_n$;

• $g(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (f_{w_0, w_1}(\mathbf{x}_n) - y_n)^2$



Surface Plot of Cost Function : MSE
by varying both w_1 and w_0



Contour plot of cost (MSE) by
varying both w_1 and w_0

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Linear Regression: Gradient Descent along the Cost Function

- Objective: Find the value of w_0 and w_1 that minimizes the cost/error function $g(\mathbf{w})$.
- Outline:
 - Start with some w_0, w_1
 - Keep changing w_0, w_1 to reduce $g(\mathbf{w})$
 - Settle at or near a minimum

• Model: $f_{w_0, w_1}(\mathbf{x}_n) = w_0 + w_1 \mathbf{x}_n$

• Cost Function : $g(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (f_{w_0, w_1}(\mathbf{x}_n) - y_n)^2$

Weight/Parameter Updating

$$w_0 = w_0 - \alpha \frac{\partial}{\partial w_0} g(\mathbf{w})$$

$$w_1 = w_1 - \alpha \frac{\partial}{\partial w_1} g(\mathbf{w})$$

Simultaneously update the weights
Repeat until convergence

α : Learning rate

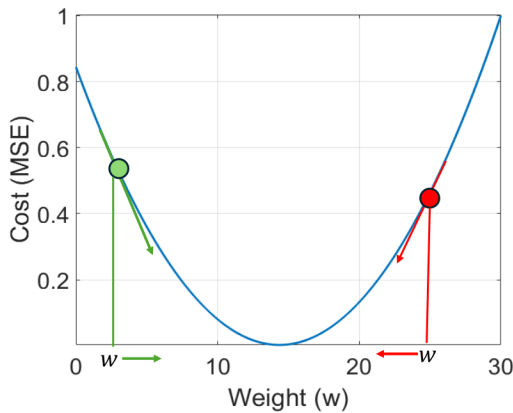
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Linear Regression: Gradient Descent along the Cost Function



$$w = w - \alpha \frac{\partial}{\partial w} g(w)$$

● $\frac{\partial}{\partial w} g(w) < 0$ (negative number)

$w = w - \alpha(\text{negative number})$: Increase w

● $\frac{\partial}{\partial w} g(w) > 0$ (positive number)

$w = w - \alpha(\text{positive number})$: Decrease w

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Linear Regression: Bias and Regression Coefficient Update

- Linear regression model: $f_{w_0, w_1}(x_n) = w_0 + w_1 x_n$
- Cost function: $g(w) = \frac{1}{N} \sum_{n=1}^N (f_{w_0, w_1}(x_n) - y_n)^2 = \frac{1}{N} \sum_{n=1}^N (w_0 + w_1 x_n - y_n)^2$

Simultaneously update the bias and regression coefficients as

$$w_0 = w_0 - \alpha \frac{\partial}{\partial w_0} g(w) = w_0 - \alpha \frac{2}{N} \sum_{n=1}^N (f_{w_0, w_1}(x_n) - y_n)$$

Bias update

$$w_1 = w_1 - \alpha \frac{\partial}{\partial w_1} g(w) = w_1 - \alpha \frac{2}{N} \sum_{n=1}^N (f_{w_0, w_1}(x_n) - y_n) x_n$$

Regression
coefficient
update

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Linear Regression: Simultaneous Updates of Bias and Regression

• Simultaneous weight update

$$w_0 = w_0 - \alpha \frac{\partial}{\partial w_0} g(\mathbf{w})$$

Correct: Simultaneously update

$$\text{temp_}w_1 = w_1 - \alpha \frac{\partial}{\partial w_1} g(\mathbf{w})$$

$$\text{temp_}w_0 = w_0 - \alpha \frac{\partial}{\partial w_0} g(\mathbf{w})$$

$$w_1 = \text{temp_}w_1$$

$$w_0 = \text{temp_}w_0$$

$$w_1 = w_1 - \alpha \frac{\partial}{\partial w_1} g(\mathbf{w})$$

Incorrect: Sequential Update

$$\text{temp_}w_1 = w_1 - \alpha \frac{\partial}{\partial w_1} g(\mathbf{w})$$

$$w_1 = \text{temp_}w_1$$

$$\text{temp_}w_0 = w_0 - \alpha \frac{\partial}{\partial w_0} g(\mathbf{w})$$

$$w_0 = \text{temp_}w_0$$

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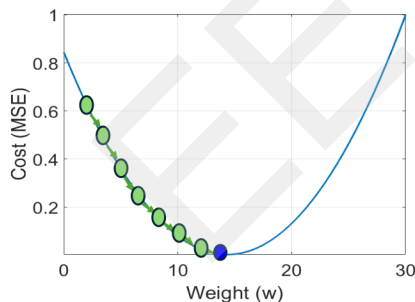
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Linear Regression: Fixing the Learning Rate

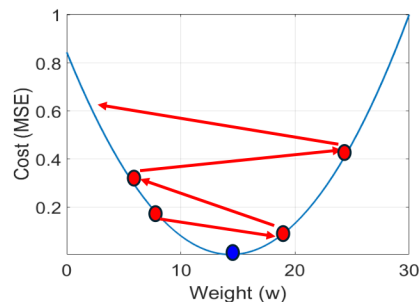
$$w = w - \alpha \frac{\partial}{\partial w} g(\mathbf{w})$$



α : is too small

Gradient descent may slow

- More iterations to converge



α : is too large

Gradient descent may:

- Overshoot, never reaches the minimum
- Fails to converge (diverge)

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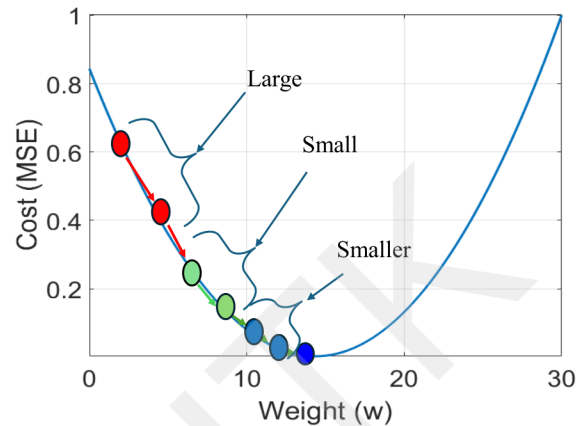
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Linear Regression: Fixed Learning Rate ?

- Can fixed learning rate reach minimum?

$$w = w - \alpha \frac{\partial}{\partial w} g(w)$$

- Near minimum
 - Derivative becomes smaller
 - Update steps become smaller
- Can reach minimum with a fixed learning rate



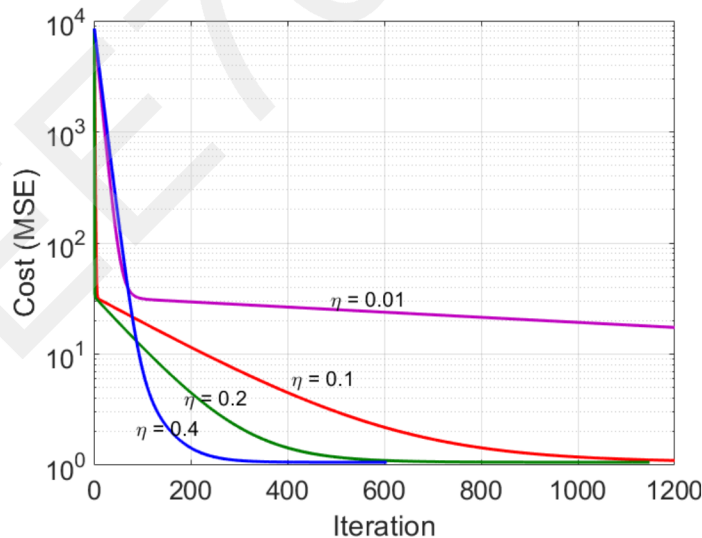
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Linear Regression: Cost Function at Various Learning Rates



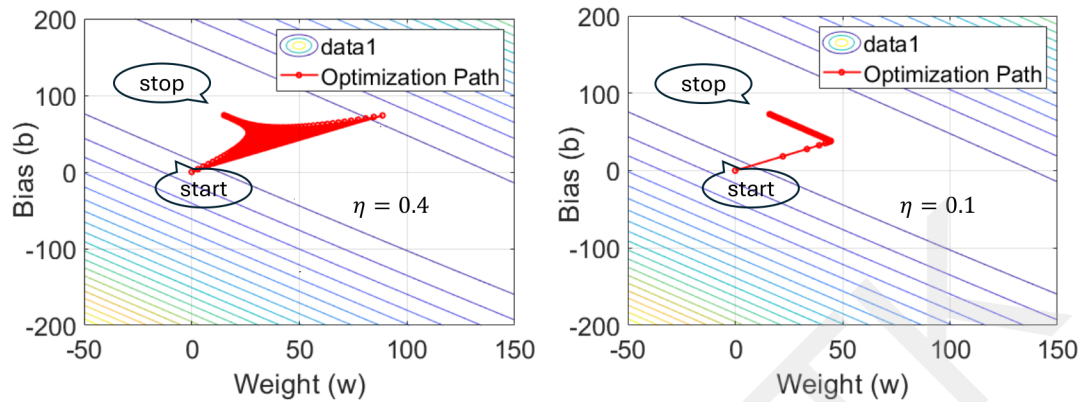
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Linear Regression: Illustrating Gradient Descent on a Contour Plot



Gradient descent path towards the minimum on the contour plot of the cost function, for various Learning rates

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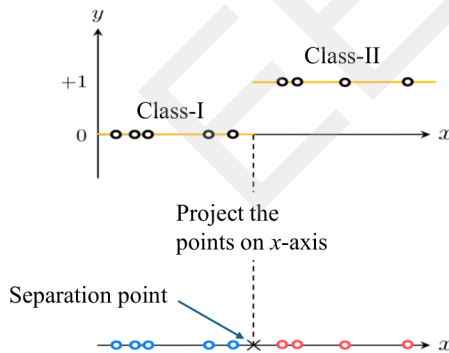
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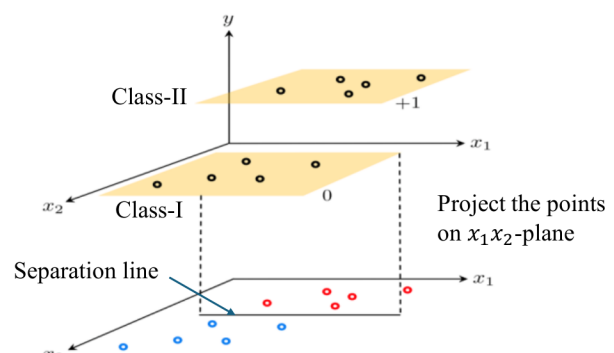
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Logistic Regression

- It is supervised machine learning algorithm used for two-class classification.
- y_n must be a categorical or discrete value, ex. $y_n \in \{0,1\}$ or $y_n \in \{True, False\}$.
- 'Steps' are the simplest shape such a dataset takes.



Classes are separated by a point when $P = 1$



Classes are separated by line when $P = 2$

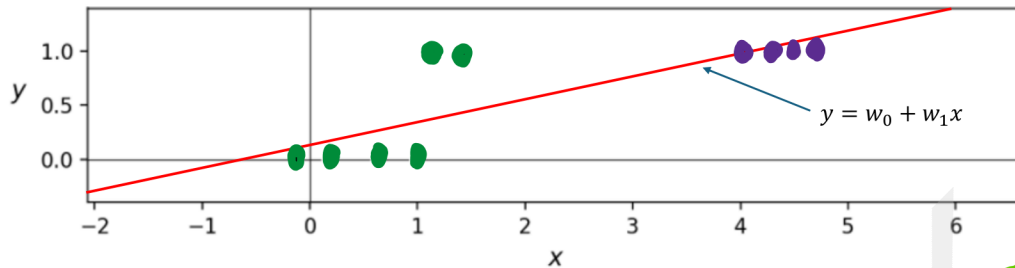
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Logistic Regression: Trying to Fit Linear Function for Irregular Data



Toy
dataset

- Fitting the line $y = w_0 + w_1 x$, results in an extremely subpar results.
- The line is inflexible to account for the nonlinearity present in such data.
- A function that matches the general shape of data, best fits.
- In other words, such data needs to be fit with a *step function*.

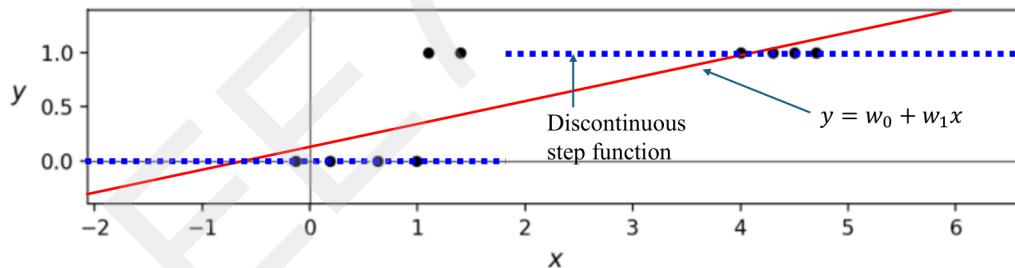
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Logistic Regression: Trying to Fit a Discontinuous Step Function



- Fit a discontinuous step function with a linear boundary.
- A linear model defining this boundary is $\text{step}(w_0 + w_1 x)$

where the step function is defined as

$$\text{step}(x) = \begin{cases} 1, & x \geq 0.5 \\ 0, & x < 0.5 \end{cases}$$

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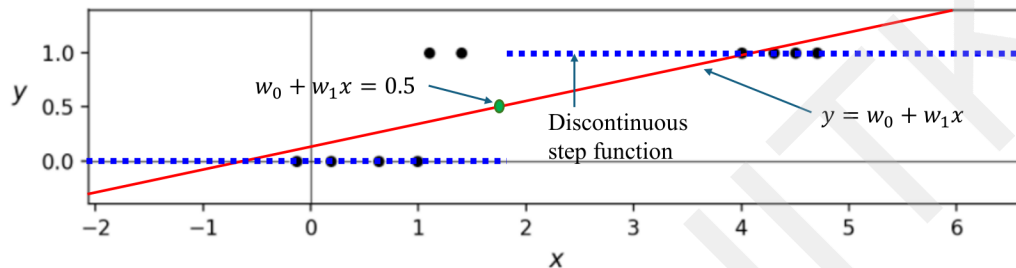
Logistic Regression: Trying to Fit a Discontinuous Step Function

Linear Boundary

- The linear boundary between the two steps is defined by all points x , where

$$w_0 + w_1x = 0.5$$
- With P dimensional input, the linear model defining the boundary as

$$\dot{\mathbf{x}}^T \mathbf{w} = w_0 + w_1x_1 + \cdots w_Px_P = 0.5$$
- Linear boundary is defined by all points, where $\dot{\mathbf{x}}^T \mathbf{w} = 0.5$



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Logistic Regression: Computing Weights of Discontinuous Step Function

Tuning the parameters

- w_0 and w_1 are tuned after composing the linear model with the step function.
- Least Squares cost function is setup that when minimized recovers ideal weights.
 - If $y_n = 1$, it lies in the +ve region where $\dot{\mathbf{x}}^T \mathbf{w} > 0.5 \rightarrow \text{step}(\dot{\mathbf{x}}^T \mathbf{w}) = 1$ matches label value.
 - If $y_n = 0$, it lies in the -ve region where $\dot{\mathbf{x}}^T \mathbf{w} < 0.5 \rightarrow \text{step}(\dot{\mathbf{x}}^T \mathbf{w}) = 0$ matches label value.
- In short, the evaluation matches its label value, i.e., $\text{step}(\dot{\mathbf{x}}_n^T \mathbf{w}) = y_n$
- Define Least Squares cost function as

$$g(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (\text{step}(\dot{\mathbf{x}}_n^T \mathbf{w}) - y_n)^2$$

- Weights are then computed by minimizing $g(\mathbf{w})$.**
- But differentiation of a discontinuous function like a step is a problem**

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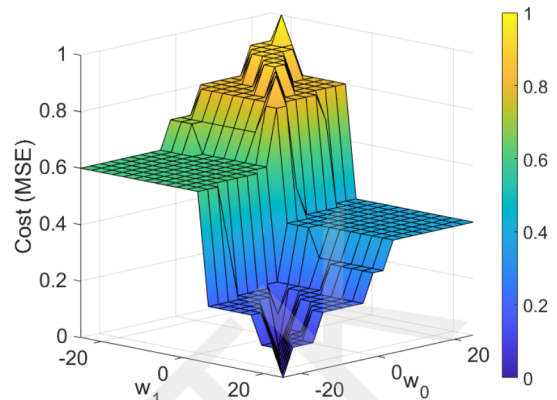
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Logistic Regression: Minimizing the Cost Function (LSE)

- $g(\mathbf{w})$ takes on only integer values.
- Impossible to minimize with local optimization, as at every point the function is completely flat.
- Local optimization cannot easily take even a single step downhill.
- This problem is inherited from the use of the step function, itself a discontinuous step.



Least Square cost function for the toy dataset

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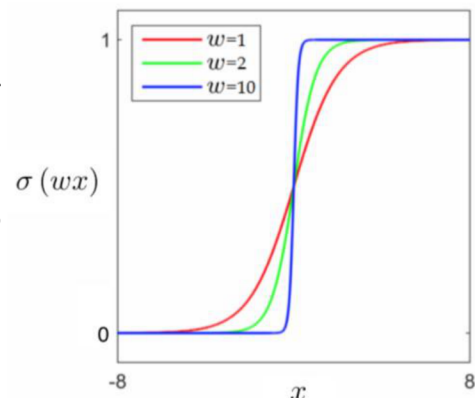
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Logistic Regression: Why Logistic Sigmoid Function

- Least Squares cannot be minimized due to the discontinuous step function.
- It needs to be replaced with a continuous function that matches it very closely everywhere.
- Such a function is the sigmoid function $\sigma(\cdot)$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

- Also called a *logistic* or *logistic sigmoid* function.
- When such a function is fit to a classification dataset, the performed regression is **Logistic regression**.



Sigmoid function

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Logistic Regression: Cost function using Logistic Sigmoid Function

- With the sigmoid function,

$$\sigma(\mathbf{x}_n^T \mathbf{w}) = y_n, \quad n = 1, \dots, N$$

- Least Squares cost for logistic regression

$$g(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (\sigma(\mathbf{x}_n^T \mathbf{w}) - y_n)^2$$

- Function is generally non-convex and contains large flat regions, making it difficult to minimize properly, but not impossible.
- Standard gradient descent, normalized gradient descent and Newton's method are used.**

$$\begin{aligned} \|x\|_1 &= \sum |x_i| \\ \|x\|_2 &= \sqrt{\sum x_i^2} \\ \|x\|_p &= (\sum |x_i|^p)^{1/p} \end{aligned}$$

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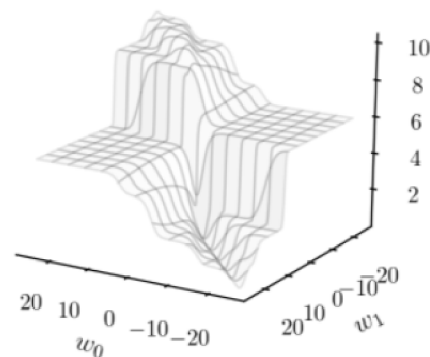
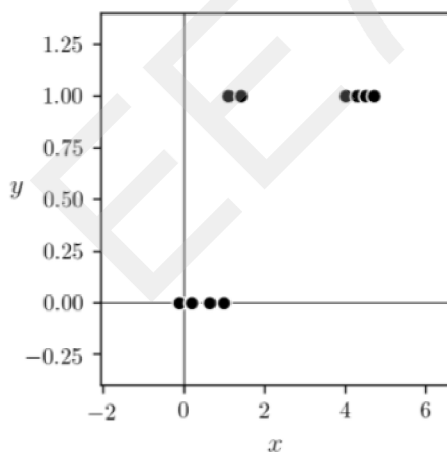
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Example: Normalized Gradient Descent for LSE Logistic Regression

- Using normalized gradient descent to minimize the logistic Least Squares cost.



Toy dataset (left) and it corresponding Least Square cost function using sigmoid function (right).

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Normalized Gradient Descent

Gradient Descent:

$$w = w - \alpha \nabla g(x)$$

- For non-convex $g(w)$, $\nabla g(x)$ vanishes near saddle points and slows the overall convergence rate and makes detection of local minima difficult.

Normalized Gradient Descent:

$$w = w - \alpha \frac{\nabla g(x)}{\|\nabla g(x)\|}$$

- $\frac{\nabla g(x)}{\|\nabla g(x)\|}$ preserves the direction of gradient but ignores magnitude.
- $\frac{\nabla g(x)}{\|\nabla g(x)\|}$ does not vanish near saddle points, so NGD does not slow down in the neighborhood of saddle points and escape “quickly”.

$\frac{y}{|y|} = \text{sign}(y)$

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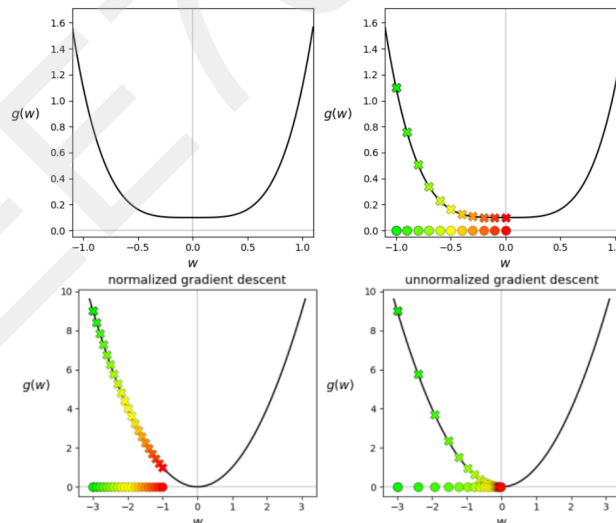
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Normalized Gradient Descent

Non-convex



Convex

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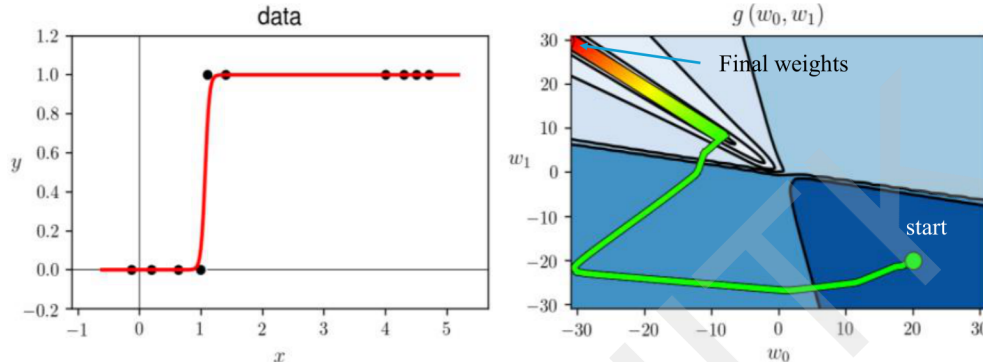
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Example: Normalized Gradient Descent for LS Logistic Regression

- Run normalized gradient descent for 900 iterations, initialized at $w_0 = -w_1 = 20$, with step length parameter fixed at $\alpha = 1$.
- The normalized gradient descent steps are colored green to red as the run progresses.
- Note : This illustration is for a **error function with several local minima**



Least Squares logistic regression fit to the data (left panel), and gradient descent path on the contour plot of the cost function (right panel).

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Logistic Regression: Logit Response Function

- Linear regression often works very well when the response variable is quantitative.
- Consider a situation where response variable takes on only two possible values, 0 and 1.
- Suppose that the model has the form

$$y_n = w_0 + w_1 x_n + \epsilon_n$$

- y_n is a Bernoulli random variable with probability distribution as:

y_n	Probability
1	$P(y_n = 1) = \pi_n$
0	$P(y_n = 0) = 1 - \pi_n$

- Since $\mathbb{E}(\epsilon_n) = 0$, the expected value of the response variable is

$$\mathbb{E}(y_n) = \sum_i y_i P(y_n = y_i) = 1(\pi_n) + 0(1 - \pi_n) = \pi_n$$

This implies $\mathbb{E}(y_n) = w_0 + w_1 x_n = \pi_n$

- This restriction can cause serious problems with the choice of the linear response function
- $$0 \leq \mathbb{E}(y_n) = \pi_n \leq 1$$

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Logistic Regression: Logit Response Function

- The expected response given by the response function $\mathbb{E}(y_n) = w_0 + w_1 x_n$ is just the probability that the response variable takes on the value 1.

- The error terms ϵ_n can only take on two values, namely,

$$\epsilon_n = \begin{cases} 1 - (w_0 + w_1 x_n), & y_n = 1 \\ -(w_0 + w_1 x_n), & y_n = 0 \end{cases} \quad \begin{matrix} 1 = w_0 + w_1 x_n + \epsilon_n \\ 0 = w_0 + w_1 x_n + \epsilon_n \end{matrix}$$

- Consequently,

- errors in this model cannot possibly be normal
- error variance is not constant.

$$\sigma_{y_i}^2 = E[y_i - E(y_i)]^2 = \pi_i(1 - \pi_i)$$

$$\sigma_{y_i}^2 = E(y_i)[1 - E(y_i)] \rightarrow \text{Fn of mean}$$

- In logistic regression, it is assumed that $\mathbb{E}(y)$ is related to x by the logit function.

$$\begin{aligned} \mathbb{E}(y) &= \frac{\exp(w_0 + w_1 x)}{1 + \exp(w_0 + w_1 x)} \\ &= \frac{1}{1 + \exp[-(w_0 + w_1 x)]} \end{aligned} \quad \text{Logit response function}$$

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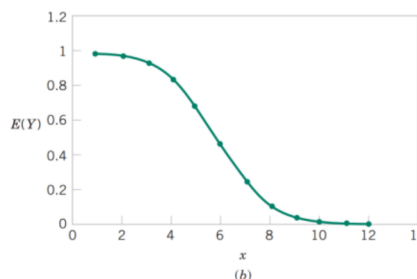
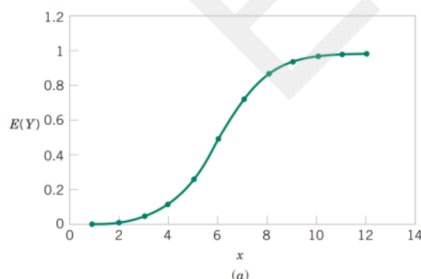
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Logistic Regression: Logit Response Function and Odds Ratio

$$\frac{\mathbb{E}(y)}{1 - \mathbb{E}(y)} = \exp(w_0 + w_1 x) \quad \text{Odds ratio}$$

- If the odds ratio is 2 for a value of x , success is twice as likely as a failure at that value.
- Estimation of parameters w_0 and w_1 is done by using MLE
- The quantity $\exp(w_0 + w_1 x)$ on the right-hand side of Equation



Examples of the logistic response function.

$$(a) \mathbb{E}(Y) = \frac{1}{1 + \exp(-6.0 - 1.0x)}$$

$$(b) \mathbb{E}(Y) = \frac{1}{1 + \exp(-6.0 + 1.0x)}$$

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Logistic Regression Using Maximum Log-Likelihood Estimation

$$\frac{\mathbb{E}(y)}{1 - \mathbb{E}(y)} = \exp(w_0 + w_1 x) \quad \text{Odds ratio}$$

$$\Pr(y = 1) = \mathbb{E}(y), \quad \Pr(y = 0) = 1 - \mathbb{E}(y)$$

$$\Pr(y) = (\mathbb{E}(y))^y (1 - \mathbb{E}(y))^{1-y}$$

$y = 1 \rightarrow \mathbb{E}(y)$
 $y = 0 \rightarrow 1 - \mathbb{E}(y)$

- Likelihood of logistic regression:

$$L(w_0, w_1 | y; x) = \prod_{n=1}^N (\mathbb{E}(y_n))^{y_n} (1 - \mathbb{E}(y_n))^{1-y_n}$$

- Log-likelihood of logistic regression:

$$l(w_0, w_1 | y; x) = \log(L(w_0, w_1 | y; x)) = \sum_{n=1}^N [y_n \log(\mathbb{E}(y_n)) + (1 - y_n) \log(1 - \mathbb{E}(y_n))]$$

- Find w_0, w_1 that maximizes the log-likelihood.

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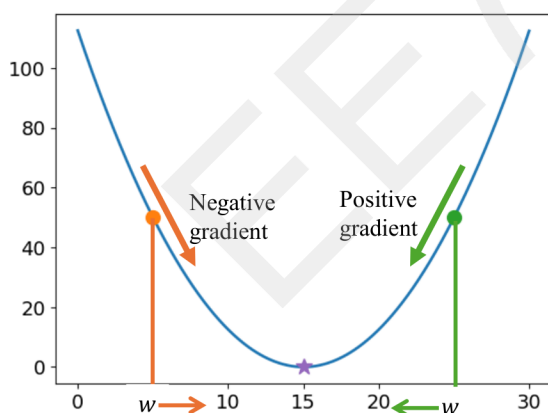
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Gradient Ascent for Maximizing Log-Likelihood

Gradient Descent: To reach minima

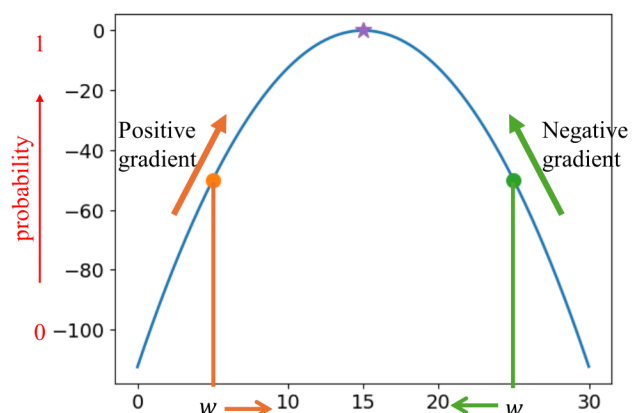


$$w^{\text{new}} = w - \alpha \frac{g(w)}{\partial w}$$

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Gradient Ascent: To reach maxima



$$w^{\text{new}} = w + \alpha \frac{g(w)}{\partial w}$$

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Logistic Regression Using Maximum Log-Likelihood Estimation

- Gradient of log-likelihood:

$$\begin{aligned}\frac{\partial l(w_0, w_1 | y; x)}{\partial w_0} &= \sum_{n=1}^N \left[y_n \frac{1}{\mathbb{E}(y_n)} \frac{\partial \mathbb{E}(y_n)}{\partial w_0} + (1 - y_n) \left(-\frac{1}{1 - \mathbb{E}(y_n)} \frac{\partial \mathbb{E}(y_n)}{\partial w_0} \right) \right] \\ &= \sum_{n=1}^N [y_n - \mathbb{E}(y_n)] \quad \leftarrow \begin{array}{l} \text{Substituting this} \\ \text{gives} \end{array} \quad \begin{array}{l} \mathbb{E}(y) = \frac{\exp(w_0 + w_1 x)}{1 + \exp(w_0 + w_1 x)} \\ \frac{\partial \mathbb{E}(y_n)}{\partial w_0} = \mathbb{E}(y_n)(1 - \mathbb{E}(y_n)) \end{array} \\ \frac{\partial l(w_0, w_1 | y; x)}{\partial w_1} &= \sum_{n=1}^N \left[y_n \frac{1}{\mathbb{E}(y_n)} \frac{\partial \mathbb{E}(y_n)}{\partial w_1} + (1 - y_n) \left(-\frac{1}{1 - \mathbb{E}(y_n)} \frac{\partial \mathbb{E}(y_n)}{\partial w_1} \right) \right] \\ &= \sum_{n=1}^N [y_n x_n - x_n \mathbb{E}(y_n)] \quad \leftarrow \begin{array}{l} \text{Substituting this} \\ \text{gives} \end{array} \quad \begin{array}{l} \frac{\partial \mathbb{E}(y_n)}{\partial w_1} = x_n \mathbb{E}(y_n)(1 - \mathbb{E}(y_n)) \end{array}\end{aligned}$$

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Logistic Regression Using Maximum Log-Likelihood Estimation

$$\mathbb{E}(y) = \frac{\exp(w_0 + w_1 x)}{1 + \exp(w_0 + w_1 x)}$$

$$\begin{aligned}\frac{\partial \mathbb{E}(y_n)}{\partial w_0} &= \frac{\left[\frac{\partial (\exp(w_0 + w_1 x))}{\partial w_0} \right] [1 + \exp(w_0 + w_1 x)] - [\exp(w_0 + w_1 x)] \left[\frac{\partial (1 + \exp(w_0 + w_1 x))}{\partial w_0} \right]}{[1 + \exp(w_0 + w_1 x)]^2} \\ &= \frac{\exp(w_0 + w_1 x) [1 + \exp(w_0 + w_1 x)] - [\exp(w_0 + w_1 x)]^2}{[1 + \exp(w_0 + w_1 x)]^2} = \frac{\exp(w_0 + w_1 x)}{[1 + \exp(w_0 + w_1 x)]^2} \\ &= \frac{\exp(w_0 + w_1 x)}{1 + \exp(w_0 + w_1 x)} \times \left[1 - \frac{\exp(w_0 + w_1 x)}{1 + \exp(w_0 + w_1 x)} \right] = \mathbb{E}(y)(1 - \mathbb{E}(y))\end{aligned}$$

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Logistic Regression Using Maximum Log-Likelihood Estimation

$$\mathbb{E}(y) = \frac{\exp(w_0 + w_1 x)}{1 + \exp(w_0 + w_1 x)}$$

$$\begin{aligned} & \frac{\partial \mathbb{E}(y_n)}{\partial w_1} \\ &= \frac{\left[\frac{\partial(\exp(w_0 + w_1 x))}{\partial w_1} \right] [1 + \exp(w_0 + w_1 x)] - [\exp(w_0 + w_1 x)] \left[\frac{\partial(1 + \exp(w_0 + w_1 x))}{\partial w_1} \right]}{[1 + \exp(w_0 + w_1 x)]^2} \\ &= \frac{x \exp(w_0 + w_1 x) [1 + \exp(w_0 + w_1 x)] - x [\exp(w_0 + w_1 x)]^2}{[1 + \exp(w_0 + w_1 x)]^2} = \frac{x \exp(w_0 + w_1 x)}{[1 + \exp(w_0 + w_1 x)]^2} \\ &= \frac{x \exp(w_0 + w_1 x)}{1 + \exp(w_0 + w_1 x)} \times \left[1 - \frac{\exp(w_0 + w_1 x)}{1 + \exp(w_0 + w_1 x)} \right] = x \mathbb{E}(y)(1 - \mathbb{E}(y)) \end{aligned}$$

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Logistic Regression Using Maximum Log-Likelihood Estimation

- The weights can be obtained using [gradient ascent](#).

$$w_0 = w_0 + \alpha \frac{\partial l(w_0, w_1 | y; x)}{\partial w_0}$$

$$w_1 = w_1 + \alpha \frac{\partial l(w_0, w_1 | y; x)}{\partial w_1}$$

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Case Study: O-ring Failure Dataset

- The dataset is on launch temperature and O-ring failure for the 24 space shuttle launches prior to the Challenger disaster of January 1986.
- There are six O-rings used on the rocket motor assembly to seal field joints.
- 1 in the "O-Ring Failure" column indicates that at least one O-ring failure had occurred on that launch.

Temperature	O-Ring Failure	Temperature	O-Ring Failure	Temperature	O-Ring Failure
53	1	68	0	75	0
56	1	69	0	75	1
57	1	70	0	76	0
63	0	70	1	76	0
66	0	70	1	78	0
67	0	70	1	79	0
67	0	72	0	80	0
67	0	73	0	81	0

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Module 3

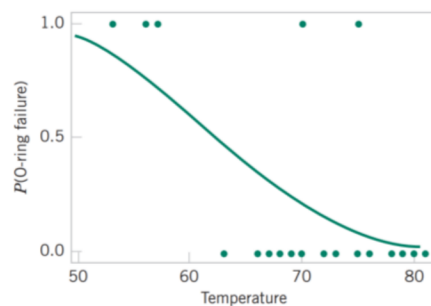
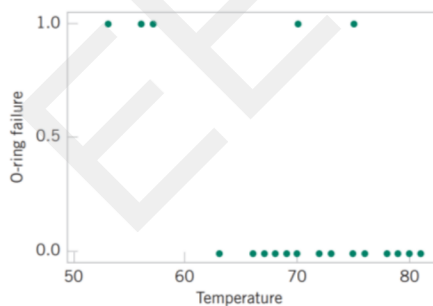
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Case Study: O-ring Failure Dataset

- The fitted logistic regression model is

$$\hat{y} = \frac{1}{1 + \exp[-(10.875 - 0.17132x)]}$$



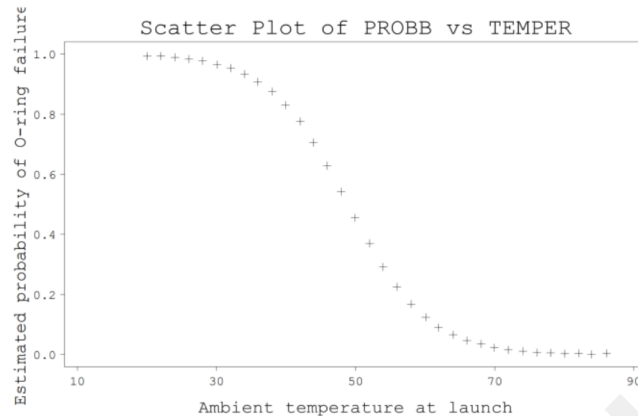
Scatter plot of O-ring failures versus launch temperature for 24 space shuttle flights. (left)
Probability of O-ring failure versus launch temperature (based on a logistic regression model). (right)

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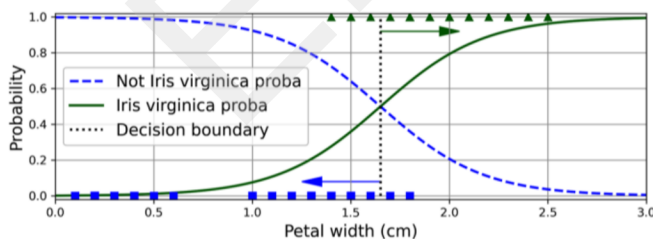


Substituting into the logistic function gives a probability estimate of O-ring failure for a temperature of 31° of **.96!** (This is an extrapolation, but you get the idea.) Indeed, with the benefit of hindsight, it can be seen that the Challenger disaster was not at all surprising, **given data that were available at the time of the flight**. As a result of its investigations, one of the recommendations of the commission was that a statistician be part of the ground control team from that time on. A complete (and more correct) discussion of this material can be found in the paper "Risk Analysis of Space Shuttle: Pre-Challenger Prediction of Failure," by S.R. Dalal, E.B. Fowlkes and B.A. Hoadley, *Journal of the American Statistical Association*, **84**, 945–957 (1989). Chapter 2 of Edward R. Tufte's 1997 book *Visual Explanations: Images and Quantities, Evidence and Narrative* discusses the background of the disaster, and the charts used by the

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Case Study: Iris Dataset

- We can use the iris dataset to illustrate logistic regression.
- A classifier is built to detect the Iris virginica type based only on the **petal width** feature.

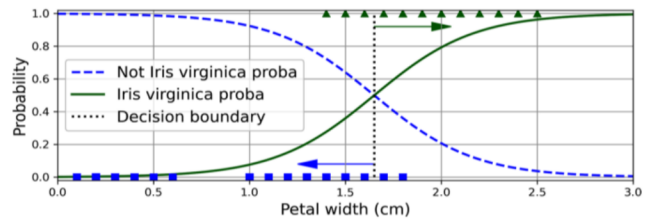


Estimated probabilities and decision boundary.



Case Study: Iris Dataset

- Petal width comparison:
 - Iris virginica flowers (in triangles) ranges from 1.4 cm to 2.5 cm,
 - Other iris flowers (in squares), ranging from 0.1 cm to 1.8 cm.
- Decision boundary:
 - Above about 2 cm the classifier is highly confident that the flower is an Iris virginica (it outputs a high probability for that class)
 - While below 1 cm it is highly confident that it is not an Iris virginica (high probability for the “Not Iris virginica” class).
 - In between these extremes, the classifier is unsure. However, the predictor will return which ever class is the most likely.
- A decision boundary is at around 1.6 cm where both probabilities are equal to 50%:
 - If the petal width is greater than 1.6 cm the classifier will predict that the flower is an Iris virginica, and otherwise it will predict that it is not.



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Case Study: Illustration of Various Decision Boundaries

- Classifier to detect the Iris virginica type based on two features: **petal width and length**.
 - Logistic regression classifier, based on two features, estimates the probability that a new flower is an Iris virginica.
-
- The dashed line represents the points where the model estimates a 50% probability: this is the model's decision boundary.
 - Each parallel line represents the points where the model outputs a specific probability, from 15% (bottom left) to 90% (top right).
 - All the flowers beyond the top-right line have over 90% chance of being Iris virginica, according to the model.

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Multiple Linear Regression: Data

$$(x_{i1}x_{i2}, \dots, x_{ik}, y_i), \quad i = 1, 2, \dots, n, \quad n > k$$

y	x_1	x_2	...	x_k
y_1	x_{11}	x_{12}	...	x_{1k}
y_2	x_{21}	x_{22}	...	x_{2k}
$\vdots$	$\vdots$	$\vdots$		$\vdots$
y_n	x_{n1}	x_{n2}	...	x_{nk}

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Multiple Linear Regression: Least Square Error

$$y_i = w_0 + w_1x_{i1} + w_2x_{i2} + \dots + w_kx_{ik} + \epsilon_i = w_0 + \sum_{j=1}^k w_jx_{ij} + \epsilon_i$$

Least square function:

$$g(\mathbf{w}) = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n \left(y_i - w_0 - \sum_{j=1}^k w_jx_{ij} \right)^2$$

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Multiple Linear Regression: Estimating Parameters

$$\frac{\partial g(\mathbf{w})}{\partial w_0} = -2 \sum_{i=1}^n \left(y_i - w_0 - \sum_{j=1}^k w_j x_{ij} \right) = 0$$

$$\frac{\partial g(\mathbf{w})}{\partial w_j} = -2 \sum_{i=1}^n \left(y_i - w_0 - \sum_{j=1}^k w_j x_{ij} \right) x_{ij} = 0, \quad j = 1, 2, \dots, k$$

Least square normal equations:

$$\begin{aligned} nw_0 + w_1 \sum_{i=1}^n x_{i1} + w_2 \sum_{i=1}^n x_{i2} + \dots + w_k \sum_{i=1}^n x_{ik} &= \sum_{i=1}^n y_i \\ w_0 \sum_{i=1}^n x_{i1} + w_1 \sum_{i=1}^n x_{i1}^2 + w_2 \sum_{i=1}^n x_{i1}x_{i2} + \dots + w_k \sum_{i=1}^n x_{i1}x_{ik} &= \sum_{i=1}^n x_{i1}y_i \\ &\vdots \\ w_0 \sum_{i=1}^n x_{ik} + w_1 \sum_{i=1}^n x_{ik}x_{i1} + w_2 \sum_{i=1}^n x_{ik}x_{i2} + \dots + w_k \sum_{i=1}^n x_{ik}^2 &= \sum_{i=1}^n x_{ik}y_i \end{aligned}$$

Can be solved using any technique appropriate and obtain parameters.

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Multiple Linear Regression: Estimating Parameters

Matrix Approach

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}$$

Cost Function:

$$\begin{aligned} g(\mathbf{w}) &= \sum_{i=1}^n \epsilon_i^2 = \boldsymbol{\epsilon}^T \boldsymbol{\epsilon} = (\mathbf{y} - \mathbf{X}\mathbf{w})^T (\mathbf{y} - \mathbf{X}\mathbf{w}) \\ &= \mathbf{y}^T \mathbf{y} - \mathbf{y}^T \mathbf{X}\mathbf{w} - \mathbf{w}^T \mathbf{X}^T \mathbf{y} + \mathbf{w}^T \mathbf{X}^T \mathbf{X}\mathbf{w} \end{aligned}$$

Normal Equation: $\frac{dg(\mathbf{w})}{d\mathbf{w}} = 0$

$$\begin{aligned} \frac{dg(\mathbf{w})}{d\mathbf{w}} &= -\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X}\mathbf{w} = 0 \\ &\Rightarrow \mathbf{X}^T \mathbf{X}\mathbf{w} = \mathbf{X}^T \mathbf{y} \end{aligned}$$

Least square estimates of parameters:

$$\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

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Multiple Linear Regression: Estimating Parameters

$$g(\mathbf{w}) = \mathbf{y}^T \mathbf{y} - \mathbf{y}^T \mathbf{X} \mathbf{w} - \mathbf{w}^T \mathbf{X}^T \mathbf{y} + \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w}$$

$$\frac{d}{d\mathbf{w}} (\mathbf{y}^T \mathbf{y}) = 0$$

$$\frac{d}{d\mathbf{w}} (\mathbf{y}^T \mathbf{X} \mathbf{w}) = (\mathbf{X}^T \mathbf{y})^T \frac{d}{d\mathbf{w}} (\mathbf{w}) = \mathbf{X}^T \mathbf{y}$$

$$\frac{d}{d\mathbf{w}} (\mathbf{w}^T \mathbf{X}^T \mathbf{y}) = \frac{d}{d\mathbf{w}} (\mathbf{w}^T) \mathbf{X}^T \mathbf{y} = \mathbf{X}^T \mathbf{y}$$

$$\frac{d}{d\mathbf{w}} (\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w}) = \frac{d}{d\mathbf{w}} (\mathbf{w}^T \mathbf{X}^T \mathbf{X}) \mathbf{w} + \mathbf{w}^T \mathbf{X}^T \mathbf{X} \frac{d}{d\mathbf{w}} (\mathbf{w})$$

$$= \frac{d}{d\mathbf{w}} (\mathbf{w}^T) \mathbf{X}^T \mathbf{X} \mathbf{w} + (\mathbf{X}^T \mathbf{X} \mathbf{w})^T \frac{d}{d\mathbf{w}} (\mathbf{w}) = \mathbf{X}^T \mathbf{X} \mathbf{w} + \mathbf{X}^T \mathbf{X} \mathbf{w} = 2\mathbf{X}^T \mathbf{X} \mathbf{w}$$

$$\frac{dg(\mathbf{w})}{d\mathbf{w}} = -\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \mathbf{w} = 0$$

$$\frac{d}{d\mathbf{w}} (\mathbf{a}^T \mathbf{w}) = \mathbf{a}$$

$$\frac{d}{d\mathbf{w}} (\mathbf{w}^T \mathbf{a}) = \mathbf{a}$$

$$\frac{d}{d\mathbf{w}} (\mathbf{w}^T \mathbf{A} \mathbf{w}) = (\mathbf{A} + \mathbf{A}^T) \mathbf{w}$$

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If $\mathbf{a}$ does not depend on $\mathbf{w}$,

$$\nabla_{\mathbf{w}} (\mathbf{a}^T \mathbf{w}) = \mathbf{a} \quad \text{and} \quad \nabla_{\mathbf{w}} (\mathbf{w}^T \mathbf{a}) = \mathbf{a}$$

So

$$\nabla_{\mathbf{w}} (-\mathbf{y}^T \mathbf{X} \mathbf{w}) = -(\mathbf{X}^T \mathbf{y}), \quad \nabla_{\mathbf{w}} (-\mathbf{w}^T \mathbf{X}^T \mathbf{y}) = -(\mathbf{X}^T \mathbf{y}).$$

(Here I used the fact that $\mathbf{y}^T \mathbf{X} \mathbf{w}$ is a scalar and can be rewritten as $(\mathbf{X}^T \mathbf{y})^T \mathbf{w}$.)

For a matrix $\mathbf{A}$ not depending on $\mathbf{w}$,

$$\nabla_{\mathbf{w}} (\mathbf{w}^T \mathbf{A} \mathbf{w}) = (\mathbf{A} + \mathbf{A}^T) \mathbf{w}$$

In your case $\mathbf{A} = \mathbf{X}^T \mathbf{X}$, and it's symmetric:

$$(\mathbf{X}^T \mathbf{X})^T = \mathbf{X}^T \mathbf{X}$$

so

$$\nabla_{\mathbf{w}} (\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w}) = 2\mathbf{X}^T \mathbf{X} \mathbf{w}.$$

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differentiate $y^T y - y^T Xw - w^T X^T y + w^T X^T Xw$
wrt w , set it to zero, and solve for

Let

$$f(w) = y^T y - y^T Xw - w^T X^T y + w^T X^T Xw.$$

1) Differentiate w.r.t. w

- $\nabla_w(y^T y) = 0$
- $\nabla_w(-y^T Xw) = -(X^T y)$
- $\nabla_w(-w^T X^T y) = -(X^T y)$
- $\nabla_w(w^T X^T Xw) = 2X^T Xw$ (since $X^T X$ is symmetric)

So

$$\nabla_w f(w) = -2X^T y + 2X^T Xw.$$

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2) Set gradient to zero

$$-2X^T y + 2X^T Xw = 0 \implies X^T Xw = X^T y.$$

3) Solve for w

If $X^T X$ is invertible,

$$w = (X^T X)^{-1} X^T y.$$

If $X^T X$ is not invertible (e.g., collinearity), the minimum-norm solution is

$$w = X^+ y$$



where X^+ is the Moore–Penrose pseudoinverse.

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Case Study: Wire Bond Data

Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2	Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2
1	9.95	2	50	14	11.66	2	360
2	24.45	8	110	15	21.65	4	205
3	31.75	11	120	16	17.89	4	400
4	35.00	10	550	17	69.00	20	600
5	25.02	8	295	18	10.30	1	585
6	16.86	4	200	19	34.93	10	540
7	14.38	2	375	20	46.59	15	250
8	9.60	2	52	21	44.88	15	290
9	24.35	9	100	22	54.12	16	510
10	27.50	8	300	23	56.63	17	590
11	17.08	4	412	24	22.13	6	100
12	37.00	11	400	25	21.15	5	400
13	41.95	12	500				

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Case Study: Wire Bond Data

Solution

$$25w_0 + 206w_1 + 8294w_2 = 725.82$$

$$206w_0 + 2396w_1 + 77177w_2 = 8008.47$$

$$8294w_0 + 77177w_1 = 3531848w_2 = 274816.71$$

$$\hat{w}_0 = 2.26379, \quad \hat{w}_1 = 2.74427, \quad \hat{w}_2 = 0.01253$$

$$\hat{y} = 2.26379 + 2.74427 + 0.01253x_2$$

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